Examiner's Use

3 **(a)** An object is placed on a smooth horizontal surface and is connected to a light spring, as shown in Fig. 3.1.

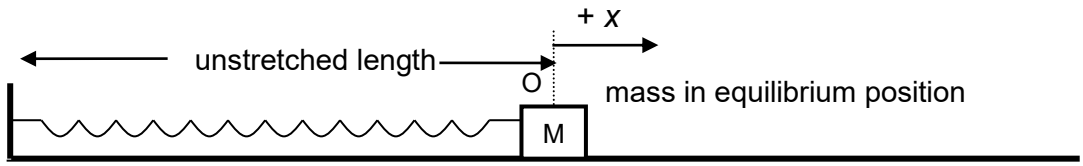

Fig. 3.1

The object is displaced to the right by 0.60 m and then released. Fig. 3.2 shows the variation with displacement *x* of acceleration *a* of the object.

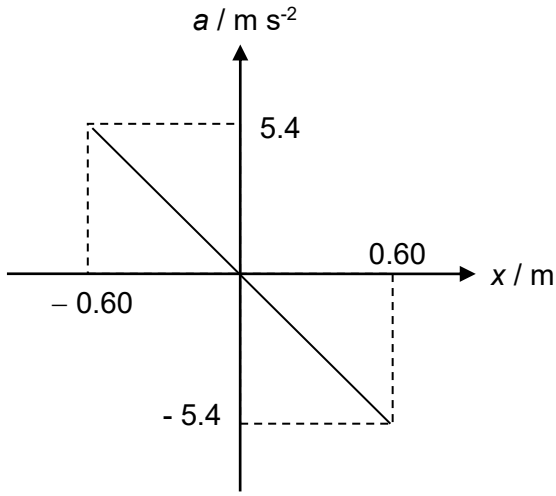

Fig. 3.2

(i) Use two features of the graph in Fig 3.2 to explain why the motion of the object is simple harmonic.

……

……

……

……

…….…… [2]

(ii) Calculate

1. the angular frequency,

angular frequency = …….. rad s^{-1} [1]

2. maximum speed of the object.

maximum speed = ……. m s^{-1} [1]

(iii) Sketch on Fig. 3.3 the variation with time of the kinetic energy of the object for one complete oscillation. The mass of the object is 0.0020 kg.

Assume that the object just passes by the equilibrium position at time = 0 s. [2]

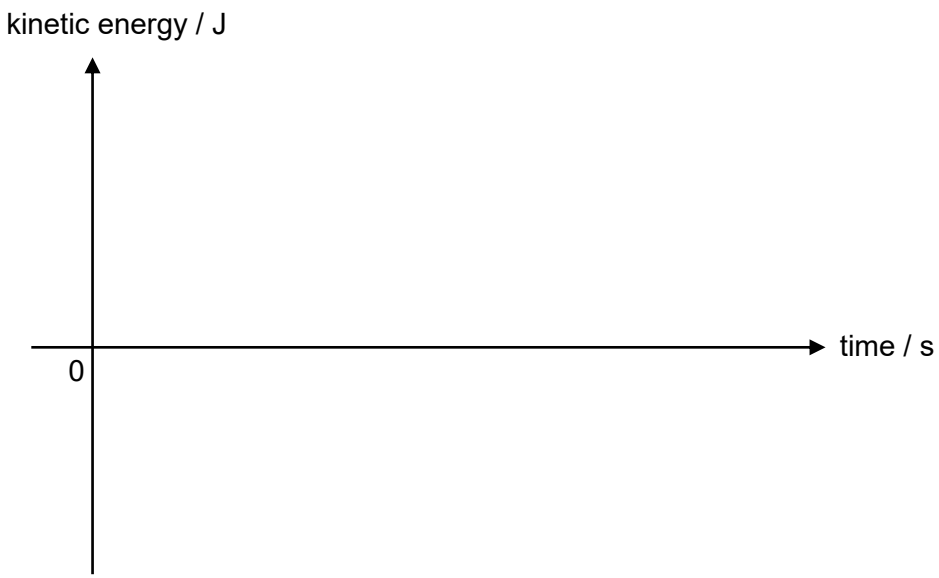

Fig. 3.3

(iv) Calculate the shortest time taken for the object to move from a point 0.30 m to the left of the equilibrium point to a point 0.30 m to the right of the equilibrium point.

time = …….. s [3]

Examiner's Use

(b) A car component of mass 0.0460 kg rattles at a resonant frequency of 35.5 Hz. Fig. 3.4 shows how the amplitude of the vertical oscillation varies with frequency.

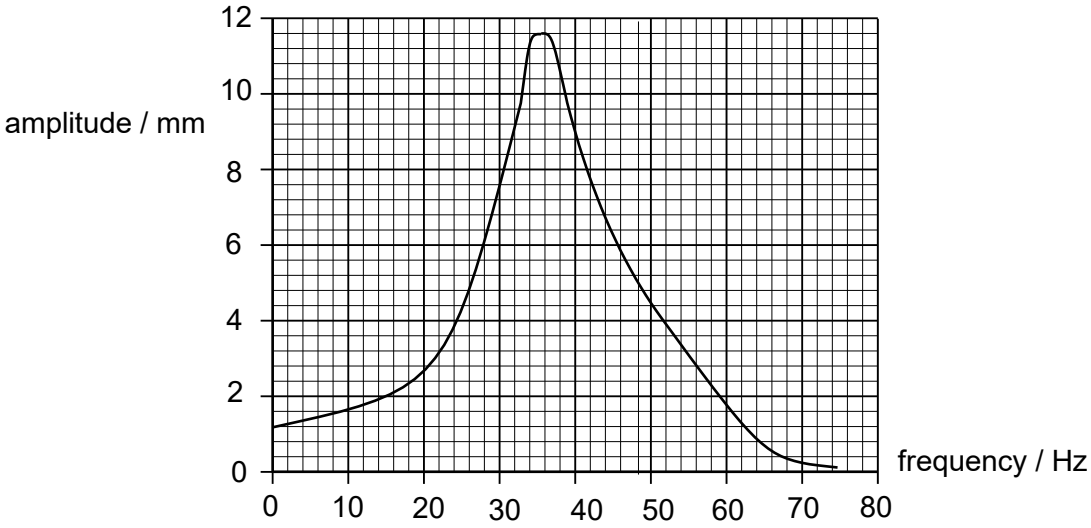

Fig. 3.4

(i) When oscillating at the resonant frequency, calculate

1. the angular frequency of the oscillation,

angular frequency = …….. rad s^{-1} [1]

2. the total energy stored in the oscillation of the component.

.

energy = ... J [2]

(ii) Draw on Fig 3.4 to show how the amplitude of the oscillation varies with frequency if the component is supported on a rubber mounting. [2]